FUNCTIONAL SPECIFICATION DOCUMENT

Zarmeen Gardens Housing Scheme — Real Estate & Society ERP

How the system sells plots, collects installments, runs post-sale operations, bills and maintains the society, controls development cost, and reports — on Frappe / ERPNext v16 with a React front end, a desk, an owner portal and WhatsApp.

PRODUCT	VERSION	DATE
Real Estate app (`realestate`)	1.0 — as built	20 October 2026
PLATFORM	**SITE**	**STATUS**
Frappe / ERPNext v16, Frappe CRM, WhatsApp	`realestate`	All phases built, acceptance 10 / 10

1. Document control

ITEM	DETAIL
Audience	Management, sales, accounts, recovery, society and project staff; implementers and testers.
Applications	Staff app (React) at `/re` · ERPNext desk (Real Estate and Society Operations workspaces) · Owner portal at `/portal` (English / اردو).
Demo data	Fictional “Zarmeen Gardens Housing Scheme”, 400 Kanal, 4 blocks, about 351 plots, 120+ bookings, regenerated with `bench --site realestate re-reset-demo` and advanced daily with `re-daily-data`.
Related	Build plan and phase log: `apps/realestate/docs/PLAN.md`. Unit tests: `realestate.tests.test_calc`, `test_geometry`.

2. Purpose and scope

The system runs a private housing scheme end to end. Every money movement posts to standard ERPNext documents (Sales Invoice, Payment Entry, Journal Entry, Purchase Invoice, Stock Entry, Asset), so the general ledger is the single source of truth and the real-estate screens add the plot, booking and society context around it.

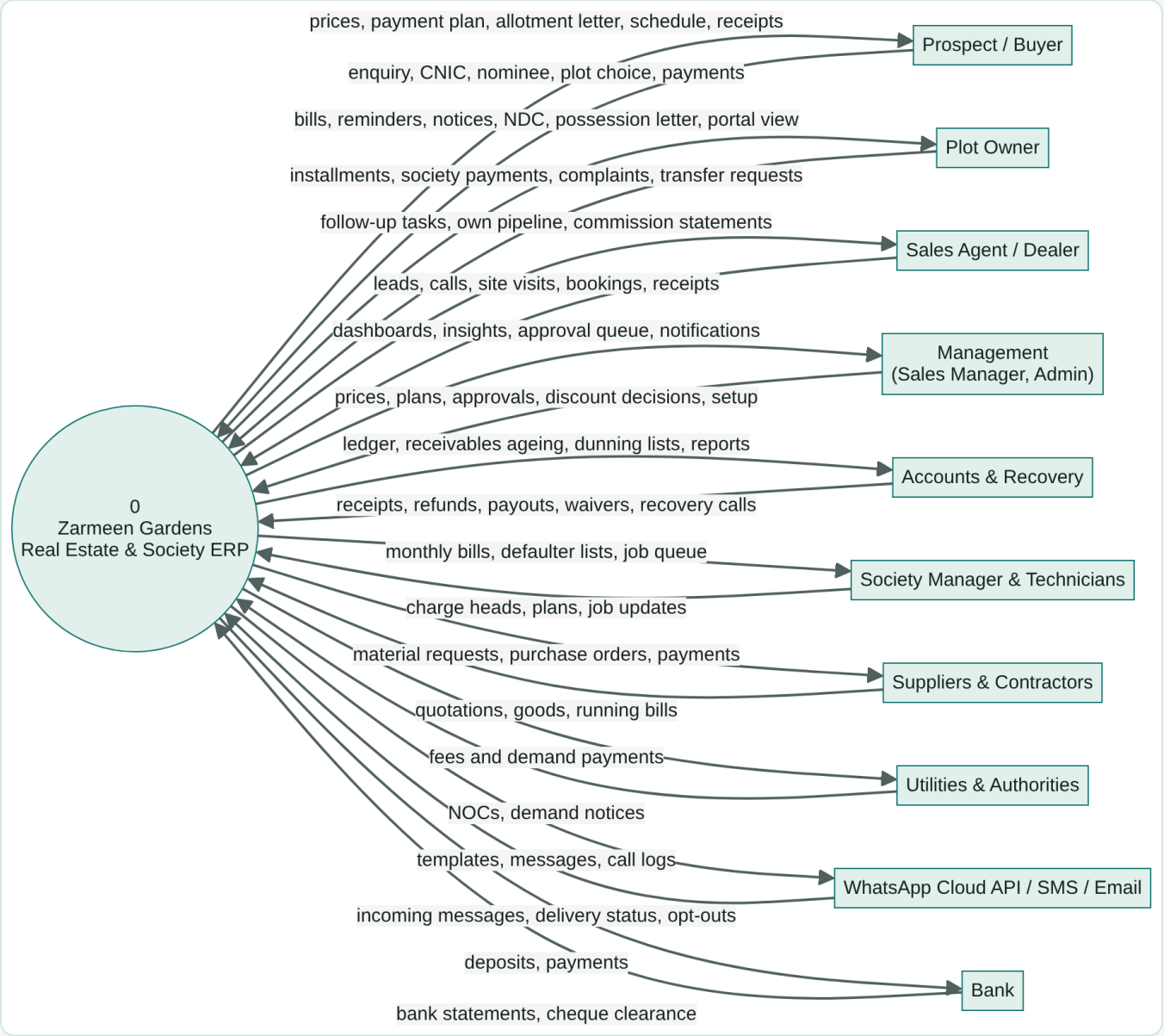
IN SCOPE	OUT OF SCOPE (THIS RELEASE)
Land and map (boundary, blocks, plots, infrastructure, KML / GeoJSON import) · CRM and WhatsApp · booking and payment plans · installments, receipts and recovery · transfers, cancellations, re-allotments, merges, surcharge waivers · agent and dealer commissions · society billing and collection · maintenance and complaints · development cost, procurement, stock and assets with depreciation · documents in English and Urdu · reports, dashboards and the owner portal.	Online card payments for owners · construction (building) management after possession · live WhatsApp Business account (runs in mock mode until credentials are supplied) · Google Maps (supported, needs an API key).

3. Users and roles

ROLE	DOES	SEES
System Manager (Admin)	Setup, settings, users, imports, acceptance checks.	Everything.
Sales Manager	Bookings, discounts above the limit, approvals for transfers, cancellations, re-allotments and merges, owner-portal invites.	All sales, CRM and collections.
Sales Agent	Leads, deals, site visits, bookings, receipts, WhatsApp chats; raises transfer / cancellation / re-allotment requests.	Only their own leads, deals, customers, bookings, receipts, visits, chats and commissions.
Accounts User / Manager	Receipts, invoices, refunds, commission payouts, supplier payments, period reports.	All financial documents.
Recovery Officer	Recovery calls, promises to pay, dunning notices, surcharge waivers (request).	Overdue bookings, society defaulters.
Society Manager	Society bills, bulk collection, charge heads, maintenance plans, infrastructure, plot merges.	Society, maintenance and map.
Projects User / Manager	Development projects, budgets, material requests, maintenance tasks.	Development and maintenance.
Plot owner (Customer)	Views plots, schedule, bills and receipts; downloads own documents; raises complaints.	Only their own records, through the portal API.

4. DFD Level 0 — context diagram

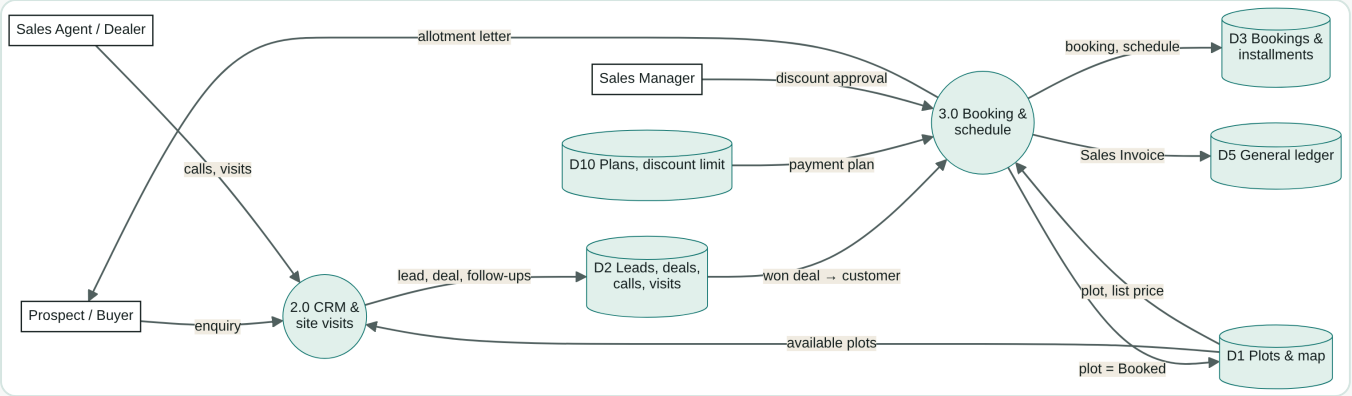
The whole system as one process and the outside parties that send data to it or receive data from it.

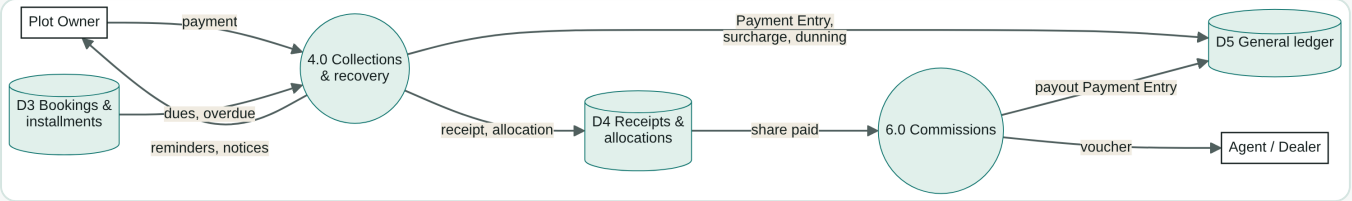


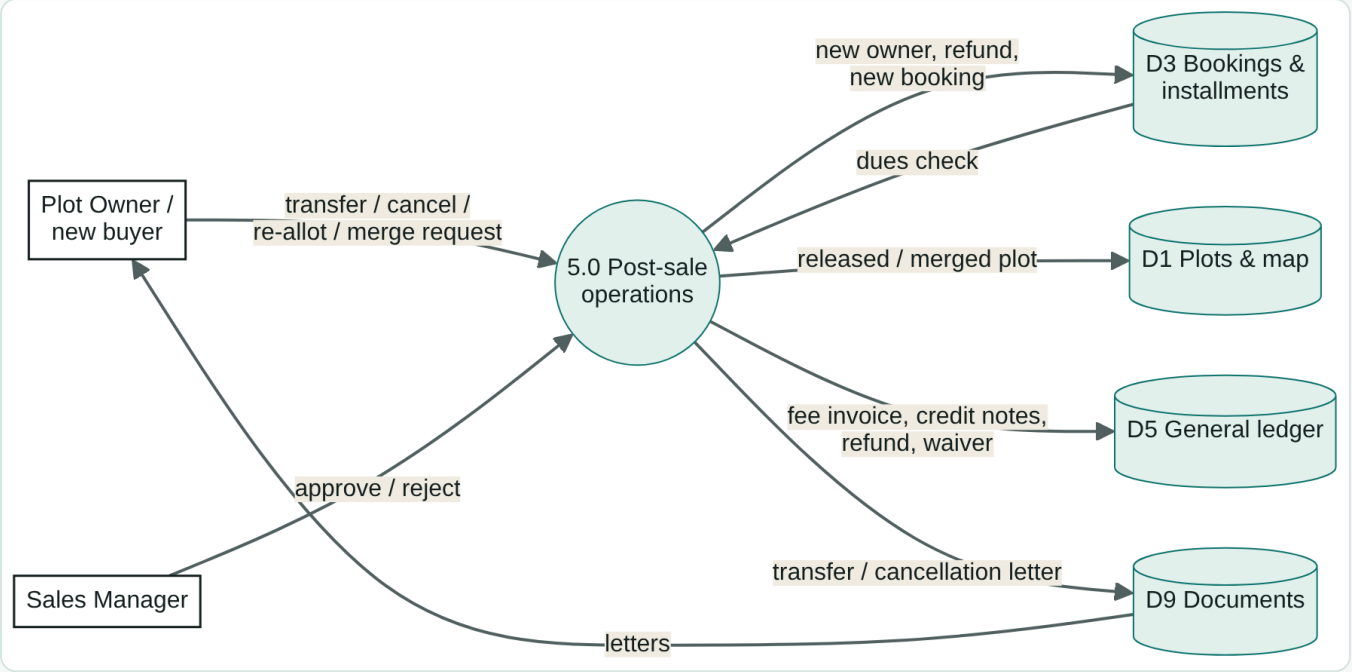
Circle the system (process 0) **Rectangle** external entity **Arrow** data flow, labelled with what moves

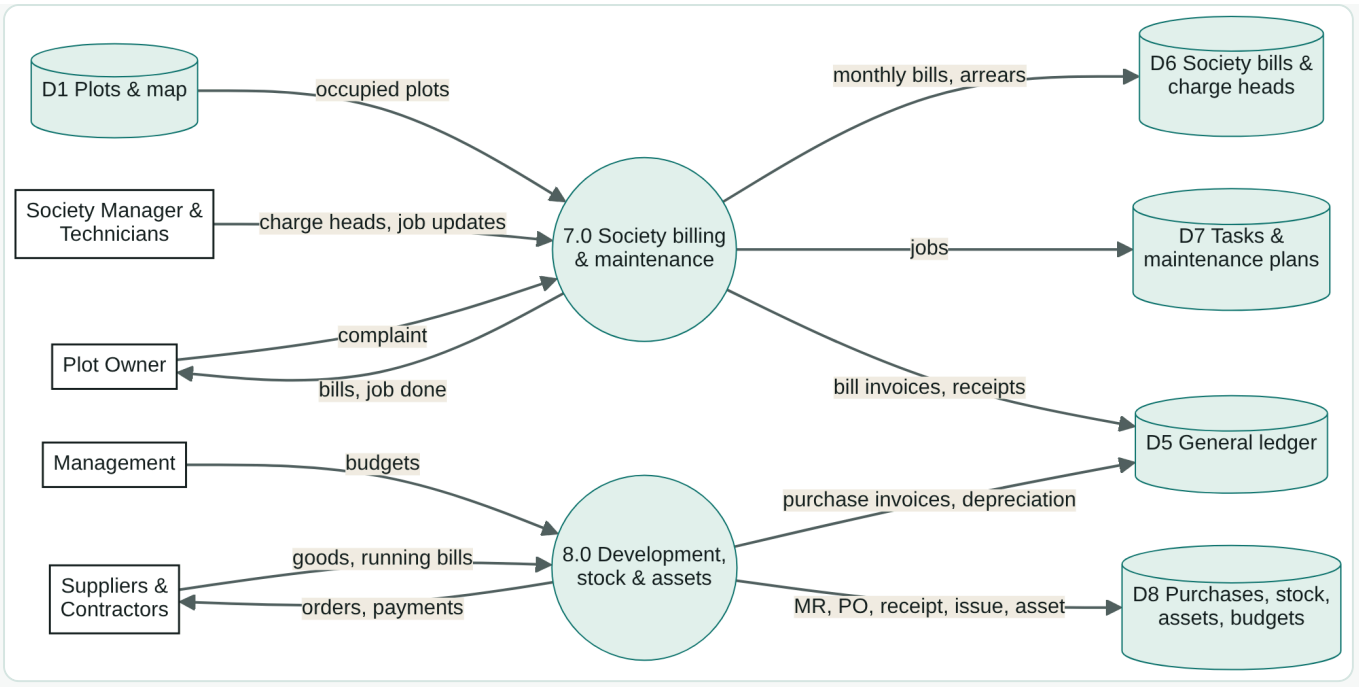
5. DFD Level 1 — main processes and data stores

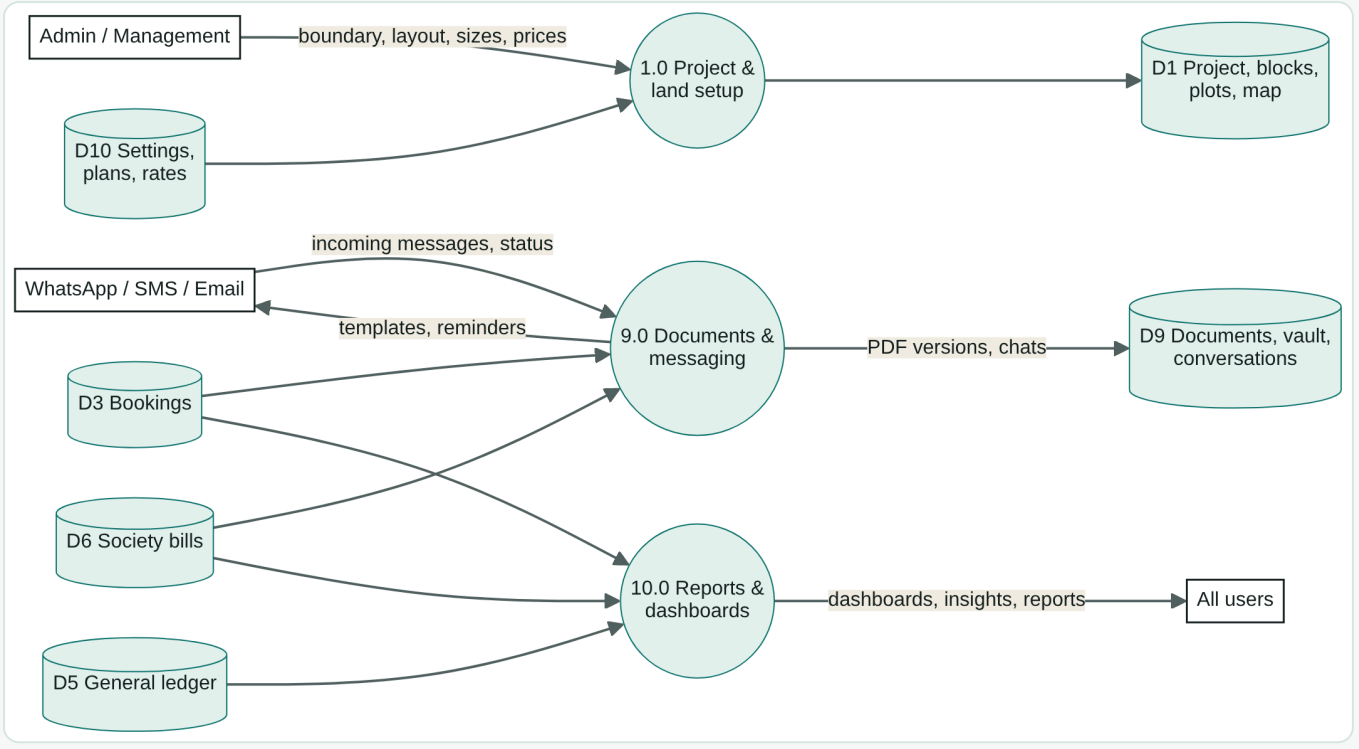
Process 0 broken into its ten main processes, the data stores they share (all in the ERPNext database) and the entities from Level 0. Stores are numbered D1–D10 and referenced in the tables below.


Sales Agent / Dealer
allotment letter
Sales Manager
discount approval
booking, schedule
D3 Bookings & installments
3.0 Booking & schedule
calls, visits
D10 Plans, discount limit
payment plan
Sales Invoice
D5 General ledger
lead, deal, follow-ups
D2 Leads, deals, calls, visits
won deal → customer
plot, list price
Prospect / Buyer
enquiry
2.0 CRM & site visits
available plots
D1 Plots & map
plot = Booked


Plot Owner
payment
4.0 Collections & recovery
Payment Entry, surcharge, dunning
D5 General ledger
D3 Bookings & installments
dues, overdue
reminders, notices
receipt, allocation
D4 Receipts & allocations
share paid
6.0 Commissions
payout Payment Entry
voucher
Agent / Dealer


new owner, refund, new booking
D3 Bookings & installments
dues check
Plot Owner / new buyer
transfer / cancel / re-allot / merge request
5.0 Post-sale operations
released / merged plot
D1 Plots & map
fee invoice, credit notes, refund, waiver
D5 General ledger
approve / reject
transfer / cancellation letter
Sales Manager
D9 Documents
letters





Circle process **Cylinder** data store **Rectangle** external entity

Level 1 processes

NO.	PROCESS	INPUTS	OUTPUTS	STORES
1.0	Project & land setup	Boundary and layout (draw or KML / GeoJSON import), blocks, plot sizes and prices, land-use shares	Plots with outlines, map features, land-use check (100% = boundary)	D1, D10
2.0	CRM & site visits	Enquiries (walk-in, dealer, Facebook, website, referral), calls, WhatsApp	Leads, deals (Opportunity → Site Visit → Negotiation → Booking), follow-up tasks, visit invitations	D2
3.0	Booking & schedule	Customer with CNIC and nominee, plot, price, discount, payment plan	Booking, installment schedule, Sales Invoice with payment schedule, allotment letter	D3, D5, D1
4.0	Collections & recovery	Payments (cash, bank transfer, cheque), promises to pay	Receipts allocated oldest-due first, Payment Entries, surcharges, dunning notices, reminders	D4, D5
5.0	Post-sale operations	Transfer, cancellation, re-allotment, merge, surcharge-waiver requests	Approved changes with their accounting: fee invoice, credit notes, refund, new invoice, merged plot	D3, D5, D1
6.0	Commissions	Bookings with agent / dealer, customer payments	Commission accrued at booking, payable after 25% paid, paid by Payment Entry with voucher	D5
7.0	Society billing & maintenance	Charge heads, occupied plots, complaints, maintenance plans	Monthly bills (with arrears), bulk collection, tasks for technicians, owner told on completion	D6, D7, D5
8.0	Development, stock & assets	Budgets per cost head, material requests, supplier goods and bills	POs, receipts into the site store, issues to projects, assets with depreciation, supplier payments	D8, D5
9.0	Documents & messaging	Booking, bill, plot, transfer and customer data	16 document types in English or Urdu (versioned PDFs), vault with expiry alerts, WhatsApp / SMS / email messages	D9
10.0	Reports & dashboards	All stores	Dashboard, insights, 21 reports (preview, chart, Excel, PDF), notifications	all

Data stores

STORE	DOCTYPES
D1	RE Project, RE Block, RE Plot (price history), RE Plot Size, RE Map Feature, RE Map Import, RE Land Use
D2	CRM Lead, CRM Deal, CRM Task, CRM Call Log, RE Site Visit
D3	RE Booking (RE Installment rows), RE Payment Plan
D4	RE Payment (RE Payment Allocation rows)
D5	Sales Invoice, Payment Entry, Journal Entry, Dunning, GL Entry, RE Surcharge, RE Commission
D6	RE Society Bill (RE Bill Charge rows), RE Charge Head (RE Charge Rate rows)
D7	Task, Project (maintenance), RE Maintenance Plan, ToDo
D8	Material Request, Purchase Order / Receipt / Invoice, Stock Entry, Asset, Asset Depreciation Schedule, Budget, Project (cost heads)
D9	RE Generated Document, RE Vault Document, RE Conversation, WhatsApp Message, RE Reminder Log, RE Broadcast
D10	RE Settings, RE Payment Plan, RE Quick Reply, approval workflows

6. Functional requirements

Each requirement names the screen in the staff app (`/re/…`) and what it posts in ERPNext. The desk offers the same documents with matching buttons.

6.1 Project & land (process 1.0)

ID	REQUIREMENT	SCREEN
FR-1.1	Create a project with its boundary (drawn or imported from KML / GeoJSON), land-use shares that must total 100% of the boundary area.	Setup › Projects, Map Import
FR-1.2	Blocks with outline, plot size and planned plot count; plots with number, size, street, corner / park / boulevard flags, outline and centre point.	Setup › Blocks, Plot Records
FR-1.3	Plot sizes with base price and monthly maintenance; price revisions kept in each plot's price history.	Setup › Plot Sizes
FR-1.4	Infrastructure on the map (roads, sewer, water, power, fibre, parks, gates, buildings) with status, completion %, contractor and linked asset.	Society Operations › Infrastructure
FR-1.5	Live society map coloured by plot status or payment health; updates when a plot changes.	Overview › Society Map

6.2 CRM & site visits (2.0)

ID	REQUIREMENT	SCREEN
FR-2.1	Leads from any source, owned by an agent; an incoming WhatsApp from an unknown number creates a lead.	CRM & WhatsApp › Leads & Deals
FR-2.2	Deal pipeline Opportunity → Site Visit → Negotiation → Booking; a deal reaching Booking creates the ERPNext Customer.	Leads & Deals (pipeline board)
FR-2.3	Calls logged with outcome, promise-to-pay amount and date, next follow-up; a missed incoming call creates a follow-up task.	Inbox › Log call; Today
FR-2.4	Site visits scheduled with a WhatsApp invitation; result recorded (completed, no show, outcome).	Site Visits, Today

6.3 Booking & schedule (3.0)

ID	REQUIREMENT	POSTING
FR-3.1	A booking needs a customer with CNIC and nominee, an available plot, price, discount and payment plan. Picking the plot fills block, size and price.	—
FR-3.2	Discounts above `max_discount_percent` (5%) wait for Sales Manager approval.	—
FR-3.3	On submit: installment schedule from the plan (booking amount, down payment, monthly / quarterly installments, balloons, development charges, possession share); plot becomes Booked.	Sales Invoice with payment schedule = installments; plan's Payment Terms Template; filled-in Terms and Conditions
FR-3.4	Allotment letter, payment schedule, application form and sale agreement generated on the letterhead, in English or Urdu.	RE Generated Document (versioned PDF)

6.4 Collections & recovery (4.0)

ID	REQUIREMENT	POSTING
FR-4.1	Receipts against a booking, society bill or surcharge (cash, bank transfer, cheque, reference); booking receipts allocated to installments oldest due first.	Payment Entry against the Sales Invoice
FR-4.2	Installment status moves with the calendar: Unpaid → Due (within 7 days) → Overdue; booking shows overdue amount and since when.	—
FR-4.3	Late-payment surcharge each month after the grace period (rules in §7).	Sales Invoice (Late Payment Surcharge income)
FR-4.4	Recovery: overdue list, recovery calls with promises, final notice, dunning.	Dunning
FR-4.5	Reminders 3 days before a due date and overdue notices every 7 days on WhatsApp, with SMS / email fallback.	RE Reminder Log

6.5 Post-sale operations (5.0) — all go through approval

ID	OPERATION	RULE	POSTING ON APPROVAL
FR-5.1	Plot transfer	No overdue installments and no society dues; transfer fee Rs 50,000.	Fee Sales Invoice; Credit Note closing the seller's balance; new Sales Invoice with the remaining installments for the buyer; transfer letter
FR-5.2	Booking cancellation	Deduction 10% of price, never more than what was paid; reason recorded.	Credit Notes, refund Payment Entry (Pay), plot released, unpaid commissions cancelled; cancellation letter
FR-5.3	Re-allotment	A plot released by an approved cancellation, to a new buyer; fee Rs 25,000.	New booking and its invoice; fee invoice
FR-5.4	Plot merge	Adjacent available plots; combined outline, area and price; source plots retired.	New plot record
FR-5.5	Surcharge waiver	Part or all of a surcharge with a reason.	Credit Note against the surcharge invoice

6.6 Commissions (6.0)

`FR-6.1` Agent 1% and dealer 2% of the net price accrue at booking (party-wise payable). `FR-6.2` A commission becomes payable when the customer has paid 25% of the price. `FR-6.3` Accounts pay one or many commissions with date, mode and reference; a Payment Entry and a printable commission voucher are produced; cancelling the payment reverses it.

6.7 Society billing & maintenance (7.0)

ID	REQUIREMENT	SCREEN / POSTING
FR-7.1	Charge heads: fixed, per marla or by plot size, for all plots or residential / commercial only.	Society Operations › Charge Heads
FR-7.2	Monthly bill per occupied plot on day 1, due in 15 days, with unpaid earlier bills carried as arrears.	Sales Invoice, one line per charge head
FR-7.3	Bulk collection screen: pick owners or a block, record many payments at once.	Society Operations › Bulk Collection
FR-7.4	Complaints from the portal, WhatsApp or staff become Tasks with priority, due date and assignee; the owner is told on WhatsApp when done.	Task in the maintenance project, ToDo
FR-7.5	Maintenance plans (daily to yearly) create tasks ahead of their due date.	Maintenance Plans

6.8 Development, stock & assets (8.0)

`FR-8.1` Each development cost head (roads, sewerage, water, electrification, street lights, boundary wall, parks, public buildings, NOCs, land) is an ERPNext Project with a Budget. `FR-8.2` Material Request → Purchase Order → Purchase Receipt into the site store → Purchase Invoice; Stock Entry (Material Issue) charges the head. `FR-8.3` Contractor running bills, NOC fees and demand notices as Purchase Invoices. `FR-8.4` Fixed assets (land, buildings, transformers, generators, machinery, tools) linked to map features, depreciated straight line by category (buildings 20 years, transformers 15, generators and machinery 10, tools 5; land not depreciated).

6.9 Documents & messaging (9.0)

`FR-9.1` Sixteen document types on the letterhead with a QR verification code, generated as numbered versions in **English or Urdu** (right-to-left): application form, allotment letter, payment schedule, installment reminder, final notice, cancellation letter, NDC, possession letter, sale agreement, site plan, construction NOC, maintenance bill, account statement, transfer application, transfer letter, payment receipt. `FR-9.2` Document vault (approvals, NOCs, registry, insurance) with expiry alerts. `FR-9.3` Shared WhatsApp inbox (mine, unassigned, unread, resolved), templates, quick replies, location pins, opt-out keywords (STOP, BAND KARO), broadcasts with a rate limit.

6.10 Reports, dashboards & portal (10.0)

`FR-10.1` Dashboard, Today (visits and follow-ups), insights, customer 360 with payer segment, society, charge-head, maintenance, infrastructure and asset insights. `FR-10.2` Report centre with preview, chart (bar, line, area, pie, donut, trend), Excel and PDF. `FR-10.3` Owner portal in

English / اردو with plots on the map, schedule, bills, receipts, own document downloads and complaints; staff can preview it as an owner, read-only.

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7. Business rules and calculations

All money maths lives in one module (`realestate/calc.py`) with unit tests; values in brackets are the current settings in RE Settings.

Installment schedule

```
net = list price − discount
fixed parts (booking, down payment, balloons, development, possession) = net ×
their % , rounded to Rs 1,000
each installment = (net − fixed parts) ÷ n , rounded to Rs 1,000 ; the last
installment absorbs the difference
Σ schedule = net   (always exact)
```

Receipt allocation

```
a receipt pays unpaid rows oldest due date first ; it may not exceed the total
outstanding
```

Late-payment surcharge

```
if days_late > grace (15):
  surcharge = max(minimum (Rs 500), overdue × rate (2% per month))  capped at
overdue × cap (10%)
one surcharge per booking per month
```

Cancellation

```
deduction = min(net × deduction % (10%), amount paid)
refund    = amount paid − deduction
```

Commission

```
commission = net × rate (agent 1%, dealer 2%)
status: Accrued → Payable when paid ≥ net × 25% → Paid
```

Society bill

```
bill = Σ charge heads for the plot (fixed | rate × marla | rate for the plot size)
payable = bill + unpaid earlier bills (arrears)
```

> Rates, fees and percentages are settings, not code. Changing them affects new documents only; posted invoices keep the values they were created with.

8. Status models

RECORD	STATUSES	MOVES WHEN
Plot	Available → Hold / Reserved → Booked → Sold → Possession Given	Hold by management; booking submitted; fully paid; possession given (needs full payment and no society dues)
Booking	Draft → Due / Overdue / Paid → Cancelled	Submit; daily dues refresh; full payment; approved cancellation
Installment	Unpaid → Due → Overdue / Partly Paid → Paid	Calendar and receipts
Approval documents	Draft → Pending Approval → Approved / Rejected (→ Revise)	Requester submits; Sales Manager (or Accounts for waivers) decides
Society bill	Unpaid → Due → Overdue / Partly Paid → Paid	Calendar and receipts
Commission	Accrued → Payable → Paid (→ Cancelled)	25% collected; payout; booking cancelled
Task (job)	Open → Working → Completed / Overdue	Technician updates; due date passes

9. Scheduled jobs

WHEN	JOB
Daily	Refresh dues and statuses · society bills on bill day · installment reminders · overdue notices · vault expiry alerts
Daily	Late-payment surcharges · recovery dunnings · commission statuses
Daily	Planned maintenance tasks · mark overdue jobs
Every minute	WhatsApp delivery statuses (mock) · SMS fallback for failed messages
Month end	Asset depreciation (ERPNext)
Demo only	A live day of demo activity (`re-daily-data`) when mock mode is on and the project is the demo

10. Reports and dashboards

AREA	REPORTS
Sales & collections	Collection Summary · Installment Dues · Plot Inventory · Recovery Promised vs Received · Sales by Block · Receivables Aging · Commission Report
Society	Society Bill Status · Society Defaulters · Society Dues by Charge Head · Society Income vs Expense · Maintenance Tasks
Development	Development Budget vs Actual · Development Cost and Profitability
CRM	Agent Activity · Agent Response Time · Lead Conversion · Leads by Source · Missed Follow-ups

11. Integrations

INTEGRATION	USE	STATE
WhatsApp Cloud API (Meta)	Templates, chat, delivery status, calls; webhook for incoming messages	Mock mode until credentials are entered in RE Settings
SMS / email	Fallback when a WhatsApp message fails	Configured channels in RE Settings
Maps	MapLibre with satellite imagery; Google Maps optional	MapLibre live; Google needs an API key
ERPNext	Accounts, buying, stock, assets, projects, dunning	Live

12. Security and audit

- Role-based access per §3; agents are scoped to their own records by server-side query conditions, not only by the screens.
- Approval workflows for transfers, cancellations (refunds), re-allotments, merges and surcharge waivers; discounts above the limit need approval.
- Submitted and cancelled financial documents cannot be deleted; corrections are cancel and amend.
- Version history on every financial and settings document; the activity chain shows each record's history as linked blocks.
- Secrets (API tokens, map keys) are stored encrypted in RE Settings; rates and fees are settings.
- The owner portal reaches only the signed-in owner's records; staff preview is read-only.

13. Non-functional requirements

AREA	REQUIREMENT
Usability	Works at phone width (390 px) in light and dark themes; staff screens in English, owner portal and documents in English or Urdu.
Performance	Lists and insight headers load in about a second on the demo data; reports preview up to 1,000 rows and export all rows.
Reliability	Daily jobs are idempotent (safe to re-run); missed demo days are caught up (up to 14).
Accuracy	Schedules always total the net price; the general ledger balances; cross-page totals audited (127 checks).
Maintainability	Standard ERPNext documents for all accounting; money maths in one tested module.

14. Project and site setup

1. **Install.** Bench with Frappe / ERPNext v16, Frappe CRM and WhatsApp; `bench get-app realestate` ; create the site; `bench --site <site> install-app realestate` (setup runs roles, workflows, print formats and workspaces).
2. **Company and accounts.** Company, fiscal year (July–June), cost centers per block, common and society; bank accounts; modes of payment (cash, bank transfer, cheque).
3. **RE Settings.** Company and default project; discount limit; surcharge, transfer, re-allotment, cancellation and commission rules; billing day and due days; WhatsApp (mock or live) and fallback channels; map provider.
4. **Project and map.** Create the RE Project and draw or import the boundary; land-use shares; blocks; plot sizes and prices; plots (import from KML / GeoJSON or create); infrastructure features.
5. **Sales setup.** Payment plans; agents (users with Sales Agent) and dealers (Sales Partners); quick replies and WhatsApp templates.
6. **Society setup.** Charge heads and size rates; maintenance plans; maintenance assignee.
7. **Development setup.** Cost-head projects and budgets; site store; suppliers; asset categories with depreciation.
8. **Users.** Staff users with roles; owner-portal invites from the customer page.
9. **Go live.** Turn on the scheduler; run `bench --site <site> re-acceptance` until every check passes.

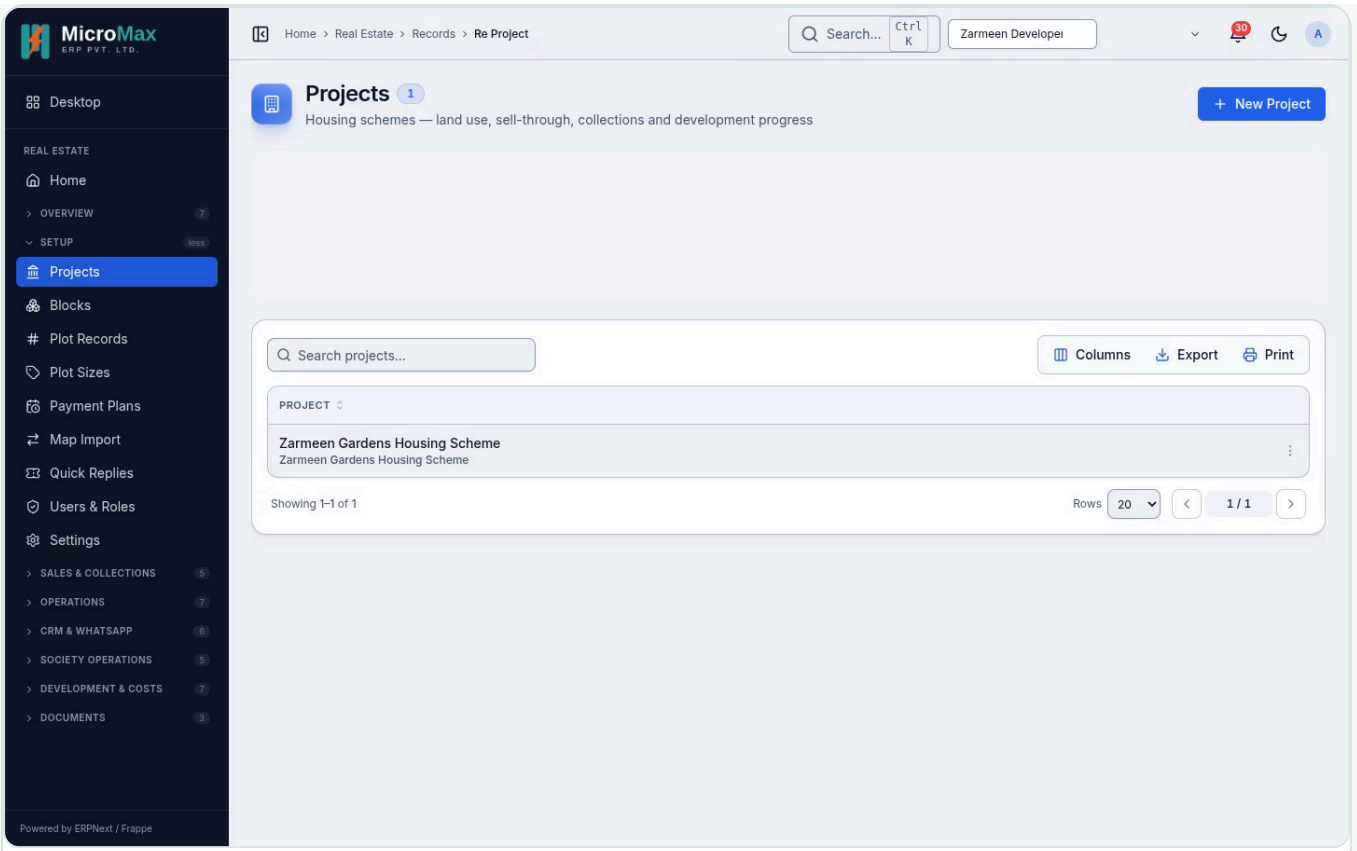
> Demo commands (`re-reset-demo` , `re-daily-data`) only act when mock mode is on and the default project is flagged as the demo; they never touch real records.

15. Acceptance criteria

NO.	CHECK	RESULT
1	Land use totals 100% of the boundary (400.00 kanal)	Pass
2	Every plot inside its block, blocks inside the boundary, no overlaps	Pass
3	Every schedule = price – discount; invoice payment schedule = invoice total	Pass
4	Every receipt has a submitted Payment Entry of the same amount; ledger balances	Pass
5	Map recolours live when a plot changes	Pass
6	Scheduled jobs enabled and current	Pass
7	WhatsApp incoming reaches the inbox and timeline	Pass (mock)
8	All documents render to PDF on the letterhead (English and Urdu)	Pass
9	Role scoping: agents, managers, portal owners	Pass
10	Audit trail and no deletion of financial records	Pass
11	Phone width, light and dark	Checked by eye

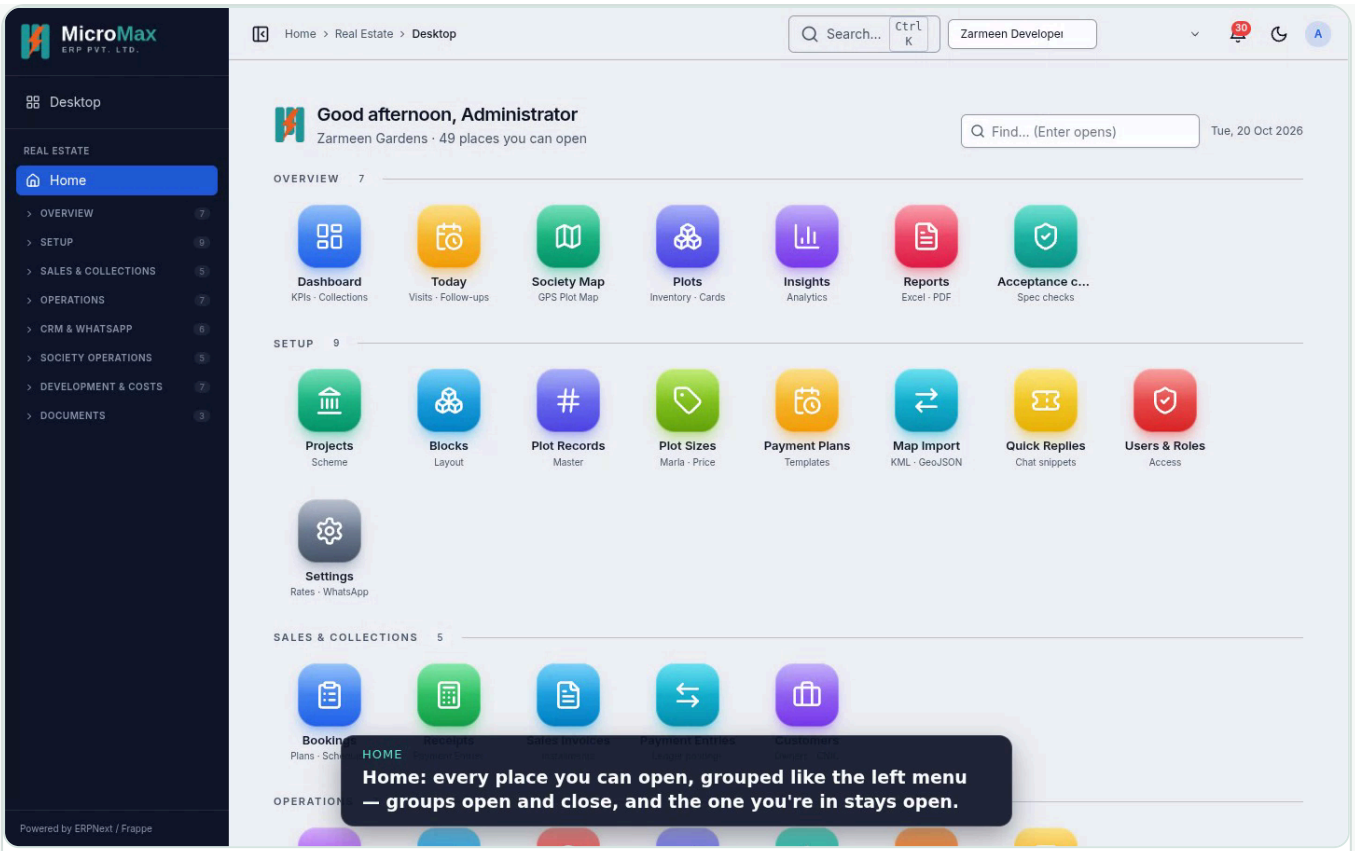
16. Video library

Screen recordings of the system on the demo scheme, one per operation. Each starts with a title card and narrates every step in a caption bar. They are re-recorded with `node tools/videos/record.mjs` in the app repository.



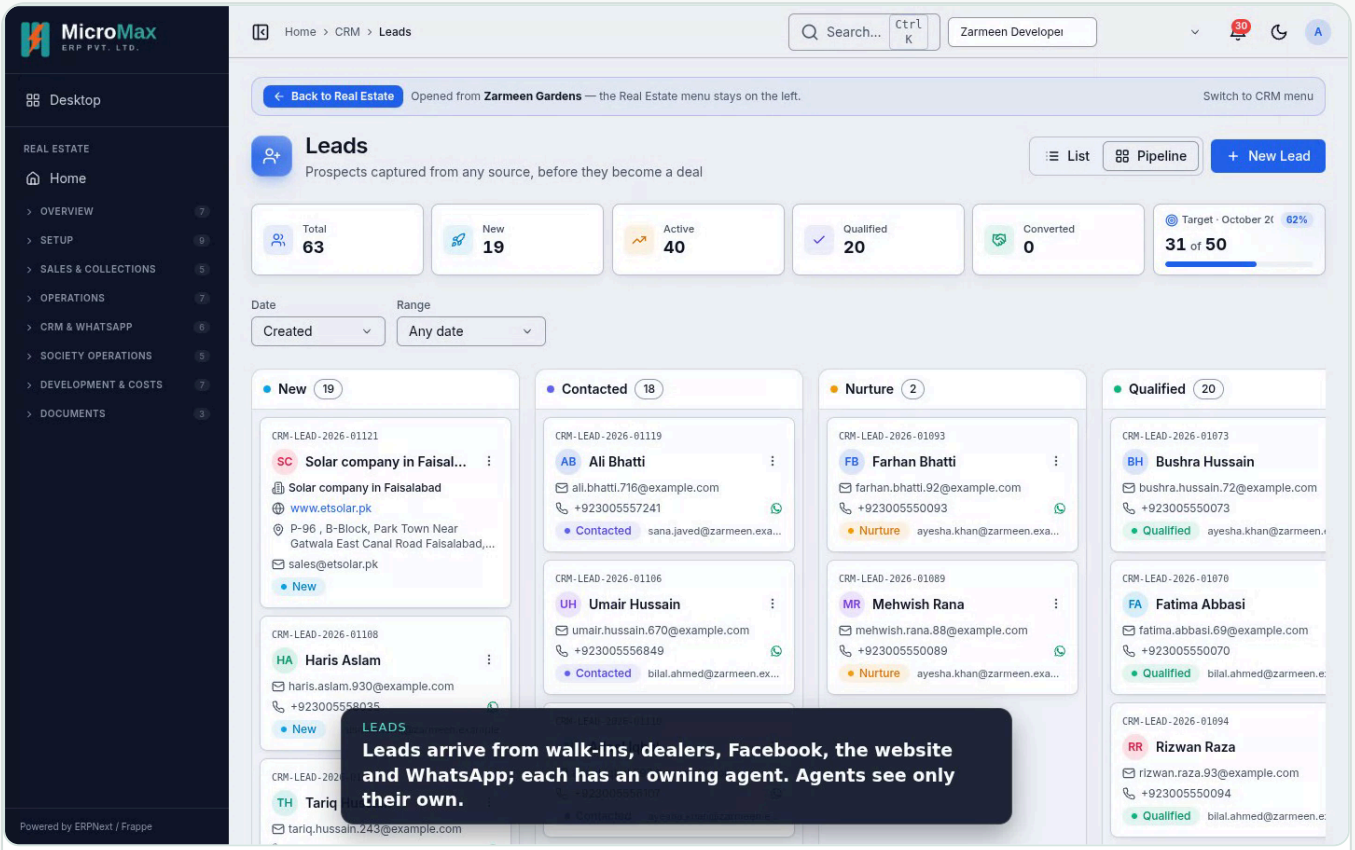
00 Project & site setup

Install, settings, project and map, blocks, plot sizes, plots, payment plans, society heads, users, acceptance checks.



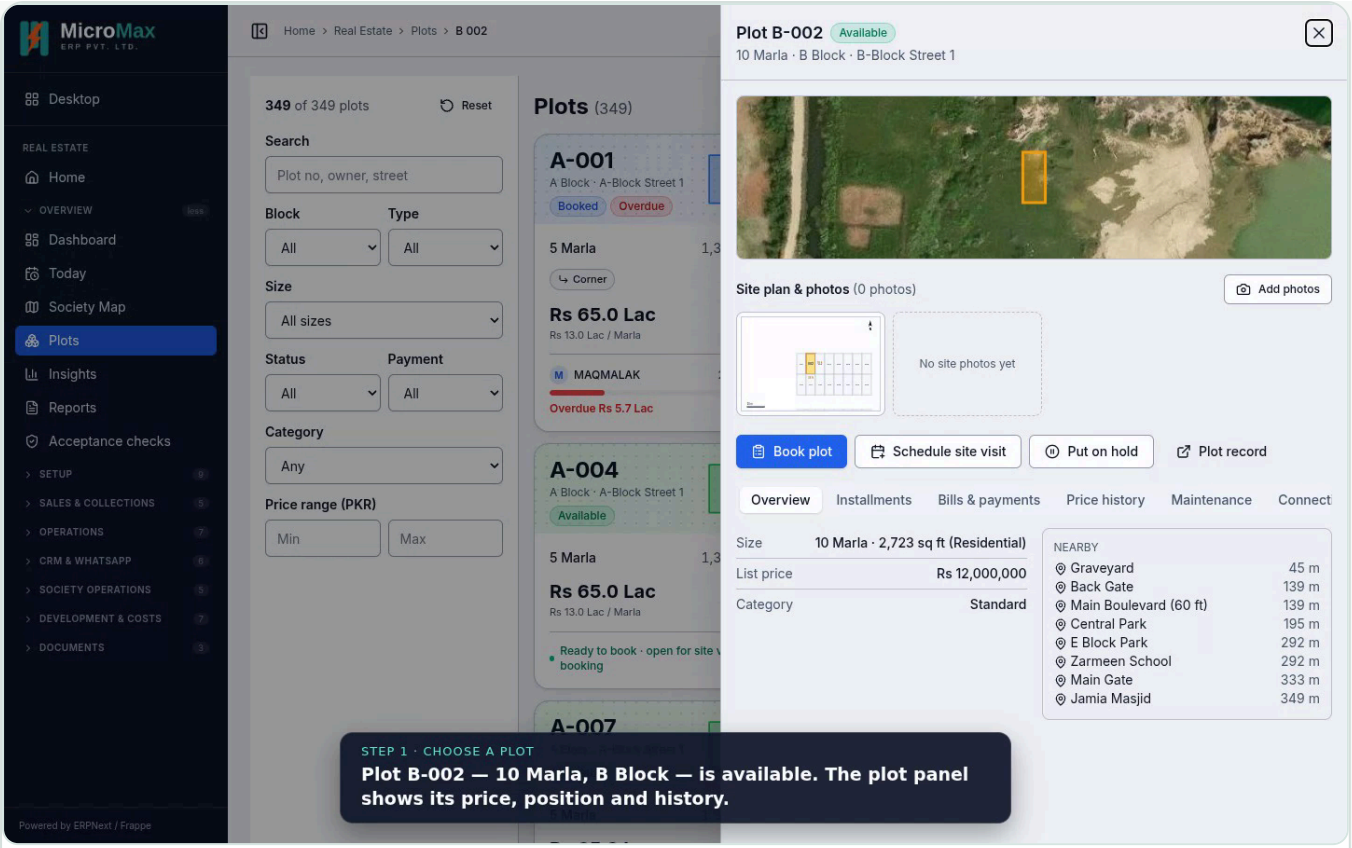
01 Your day in the app

Home, dashboard, today's visits and follow-ups, insights and notifications.



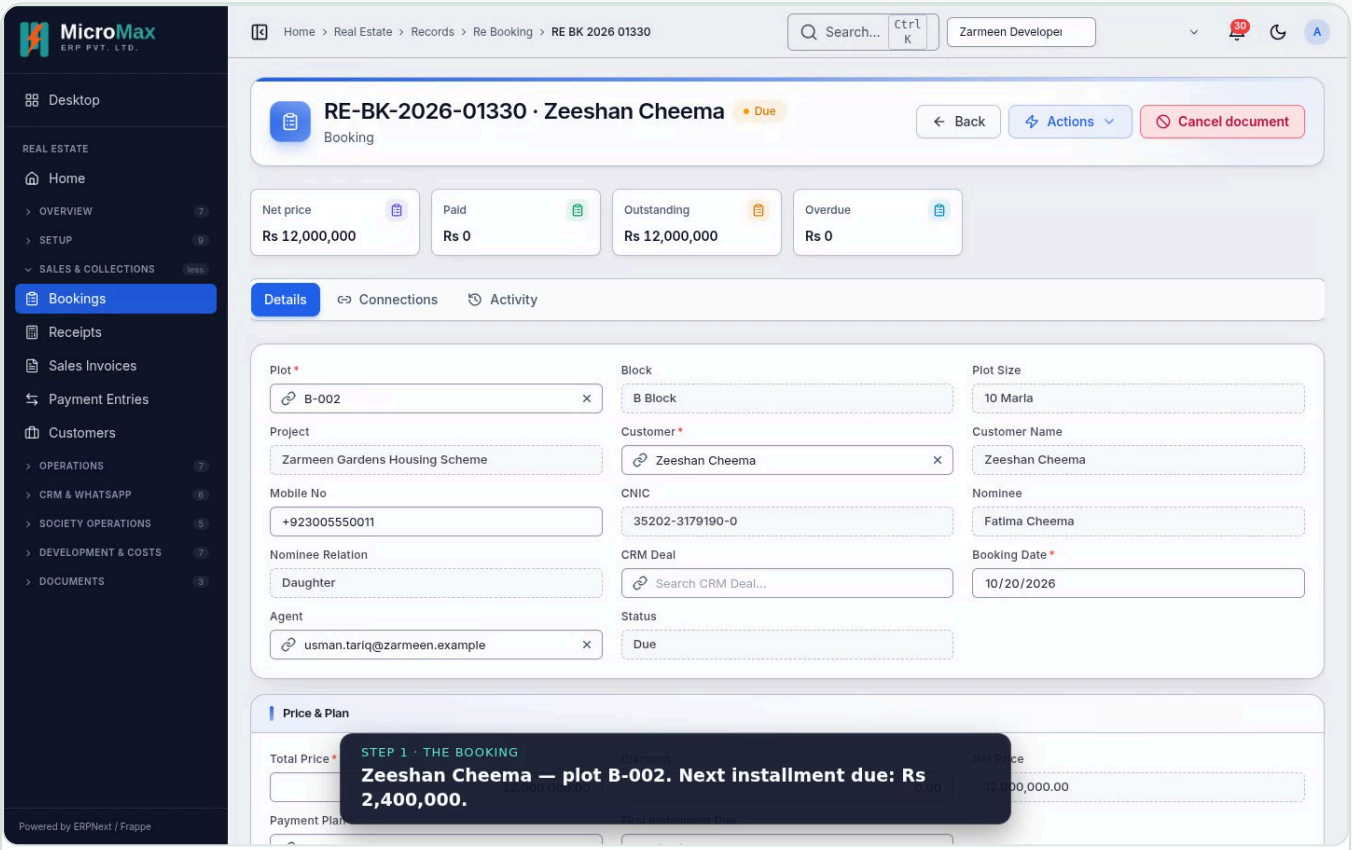
02 CRM & WhatsApp

Leads, deals, site visits and replying in the shared WhatsApp inbox.



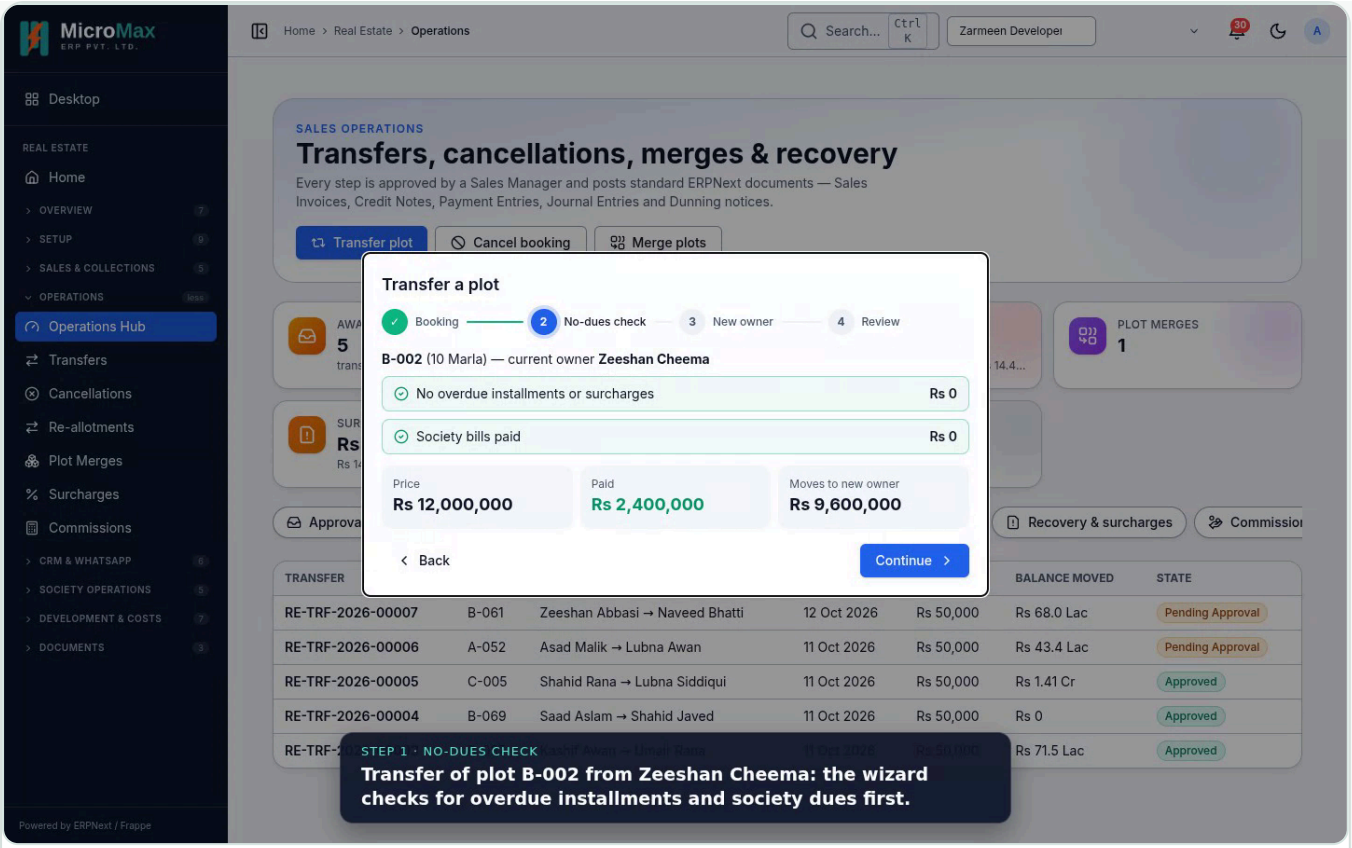
03 Booking a plot

Pick a plot, book it, submit, see the schedule and Sales Invoice, generate the allotment letter in Urdu.



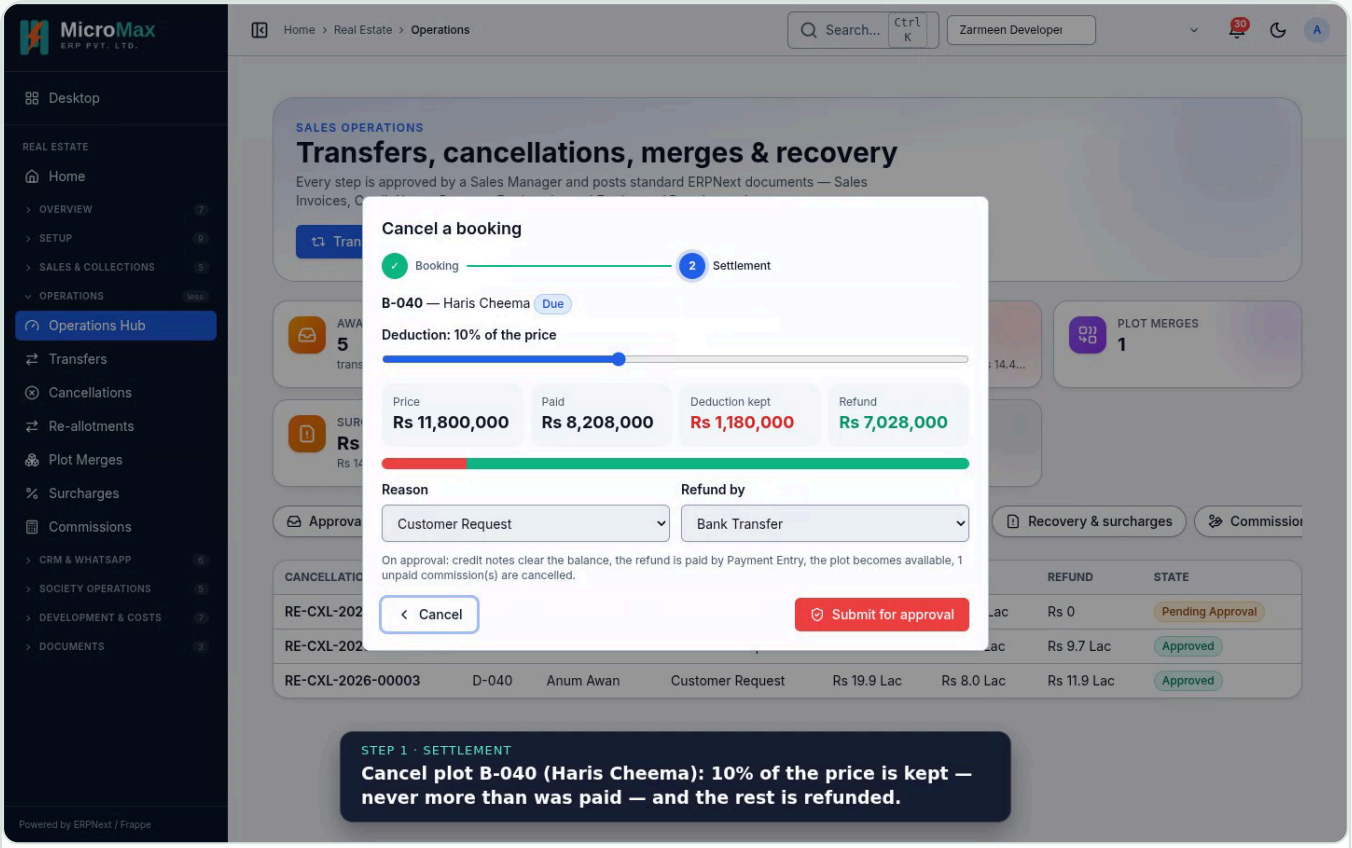
04 Receiving a payment

Receipt against a booking, oldest-due-first allocation, Payment Entry and ledger.



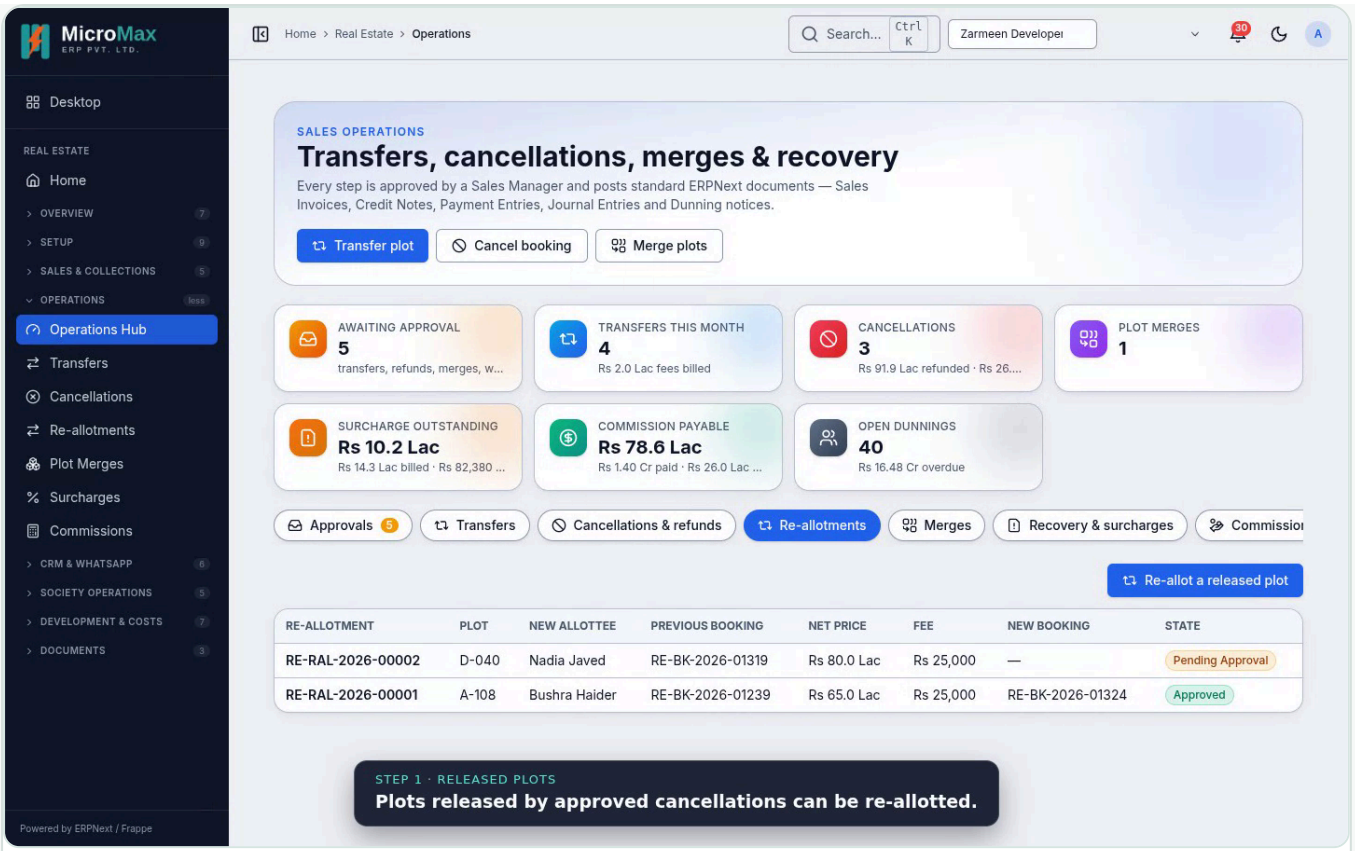
05 Transferring a plot

No-dues check, new owner, transfer fee, approval and the invoices that move.



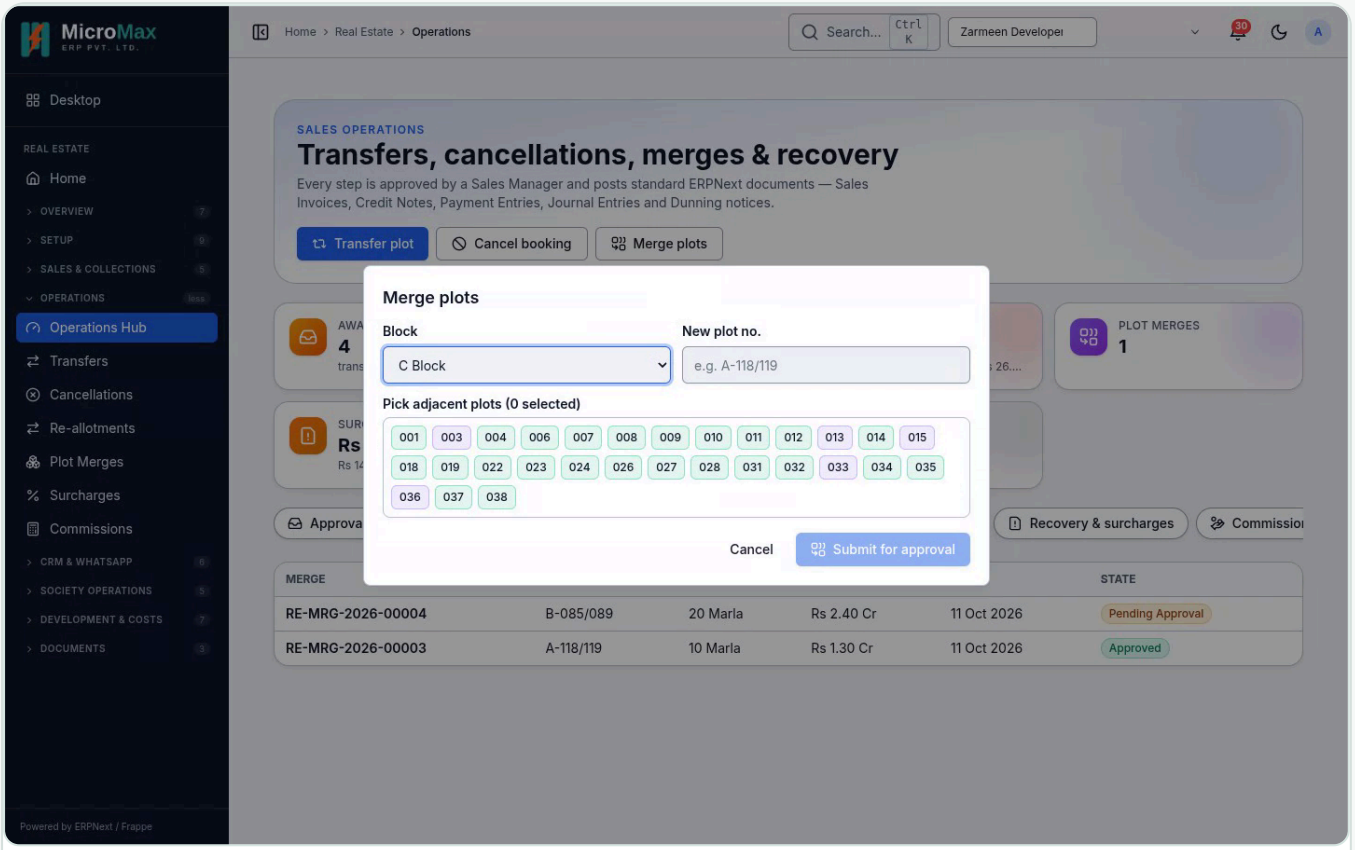
06 Cancelling a booking

Deduction and refund, approval, credit notes and the plot released.



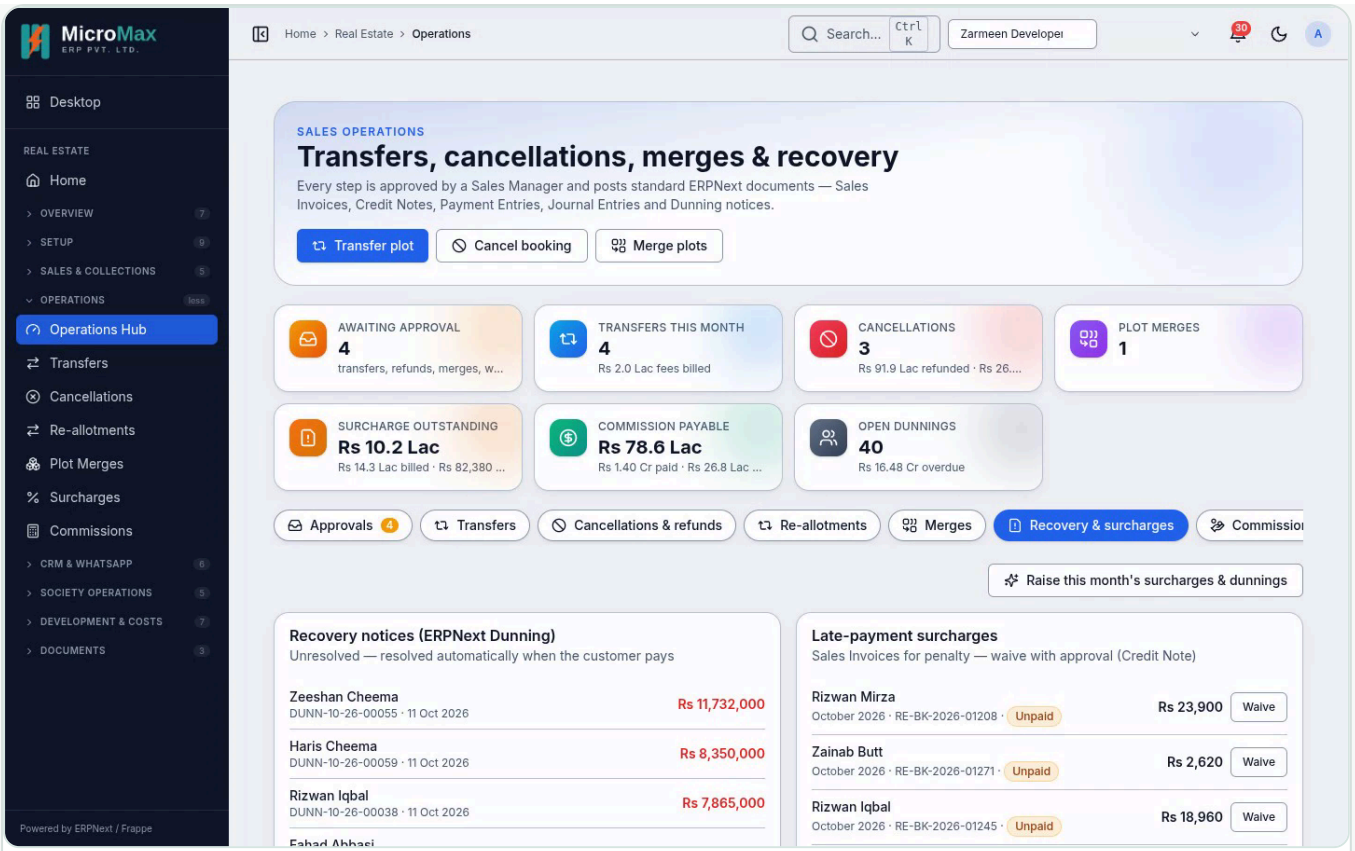
07 Re-allotting a released plot

A plot freed by a cancellation goes to a waiting buyer, with approval.



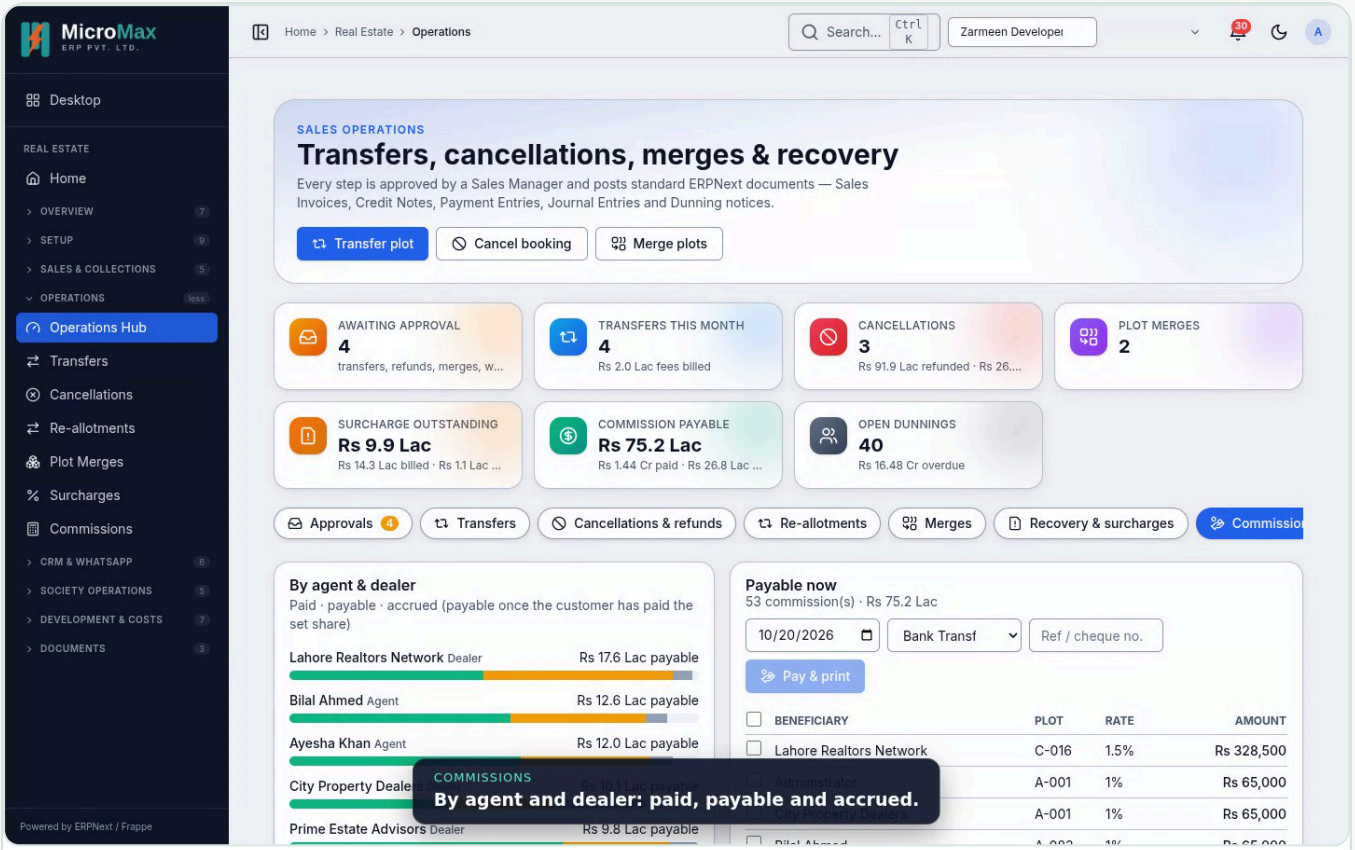
08 Merging plots

Two adjacent plots become one; approval; the map updates.



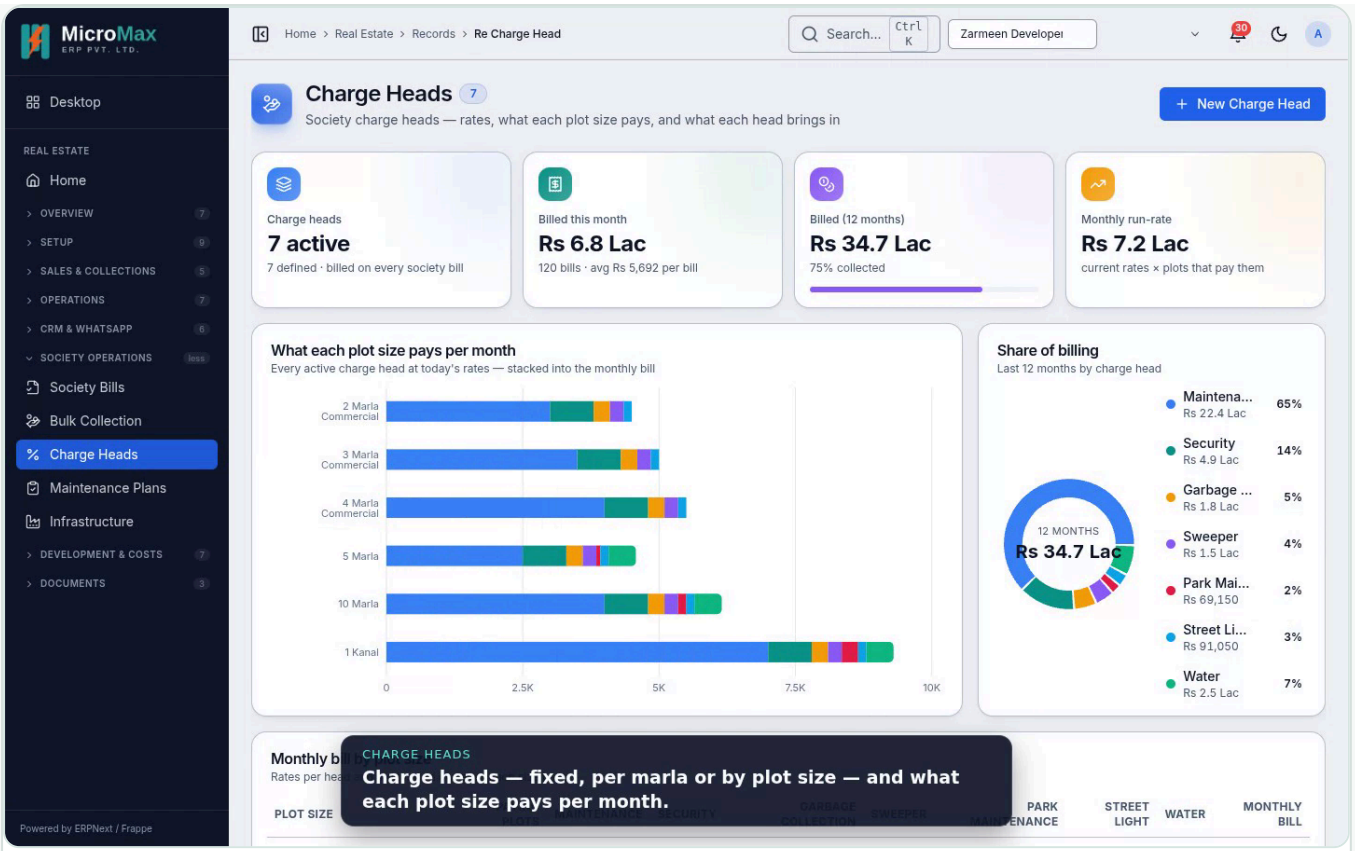
09 Recovery & surcharges

Dunning, monthly surcharges and a waiver with approval.



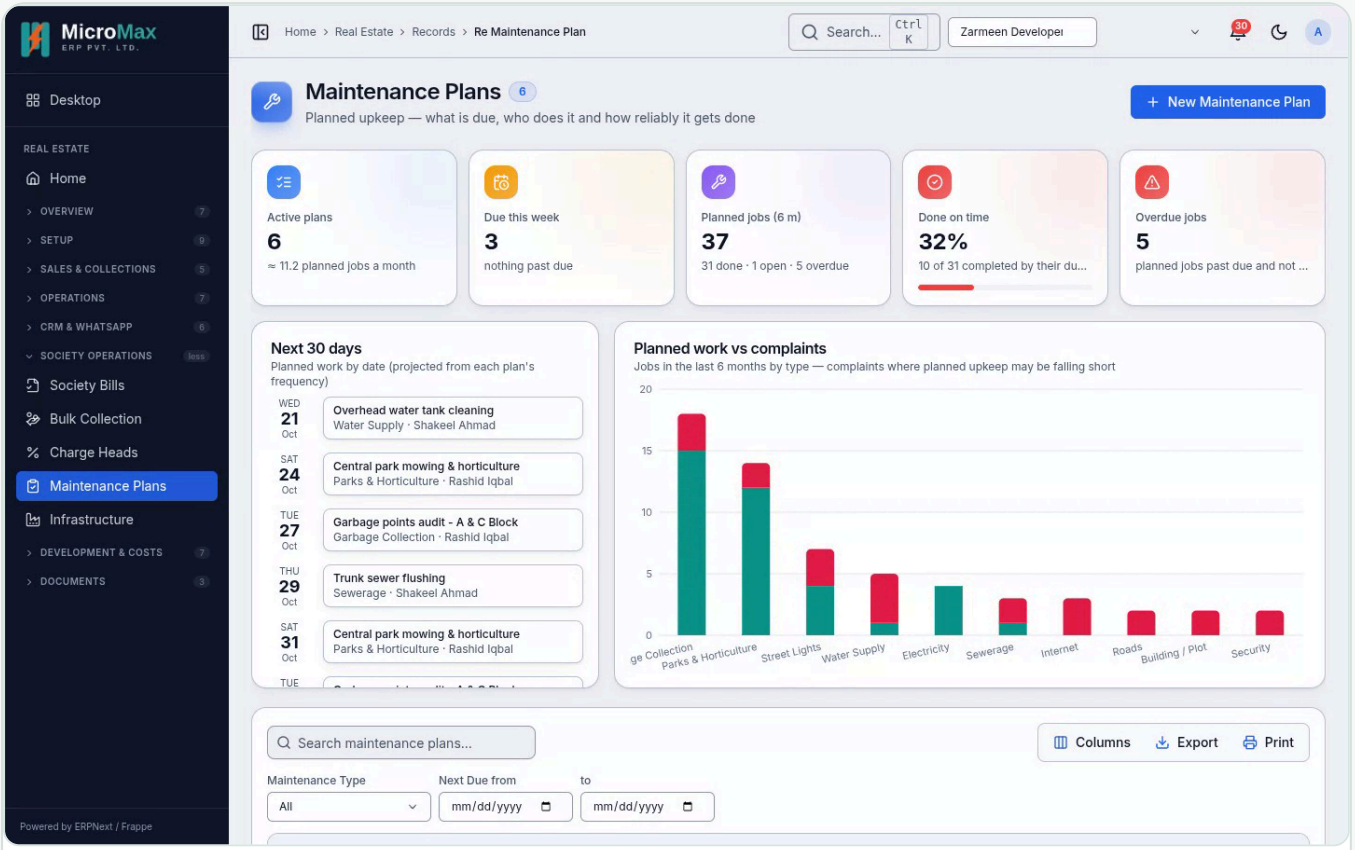
10 Commissions

Accrued, payable and paid; paying with a voucher.



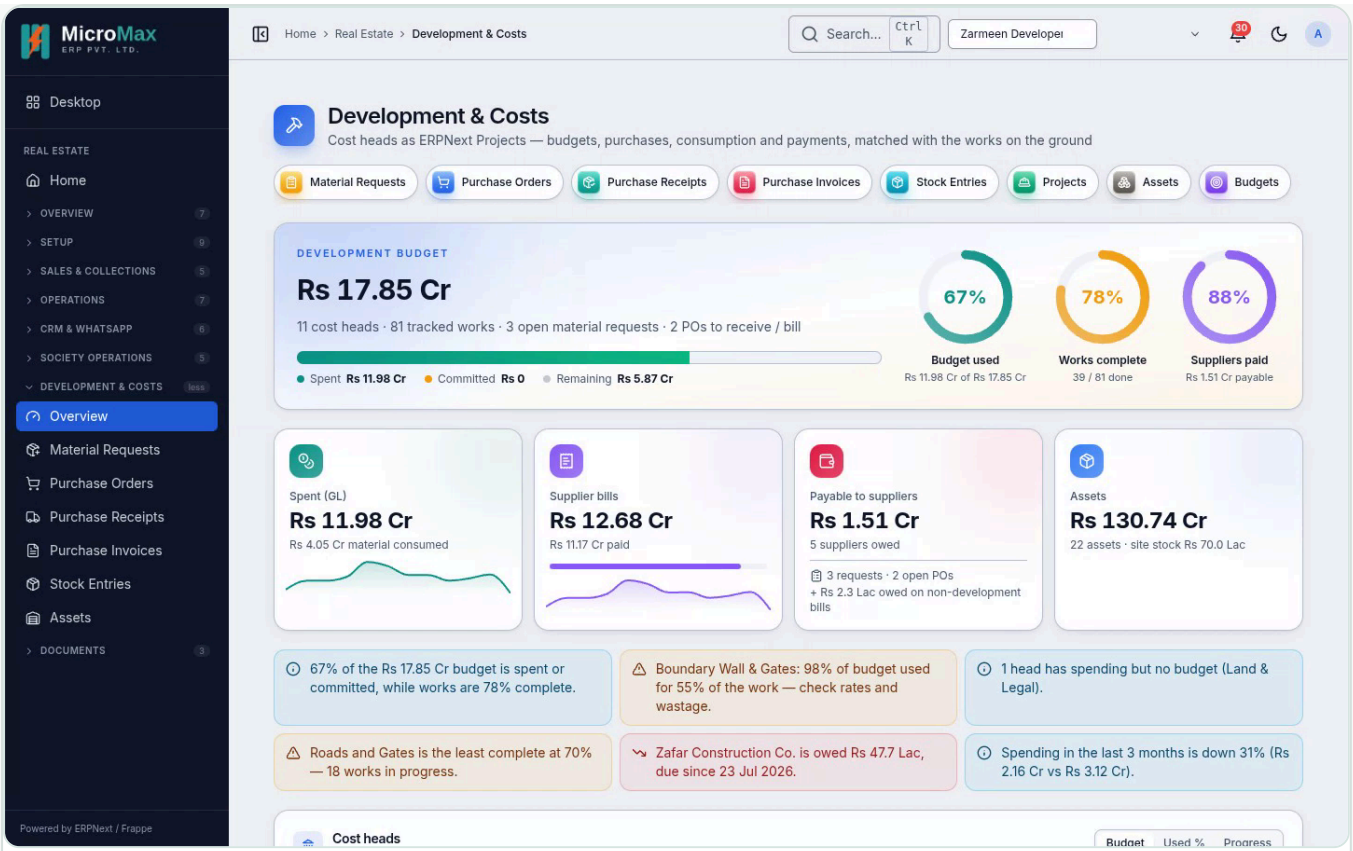
11 Society billing

Charge heads, monthly bills with arrears, insights and bulk collection.



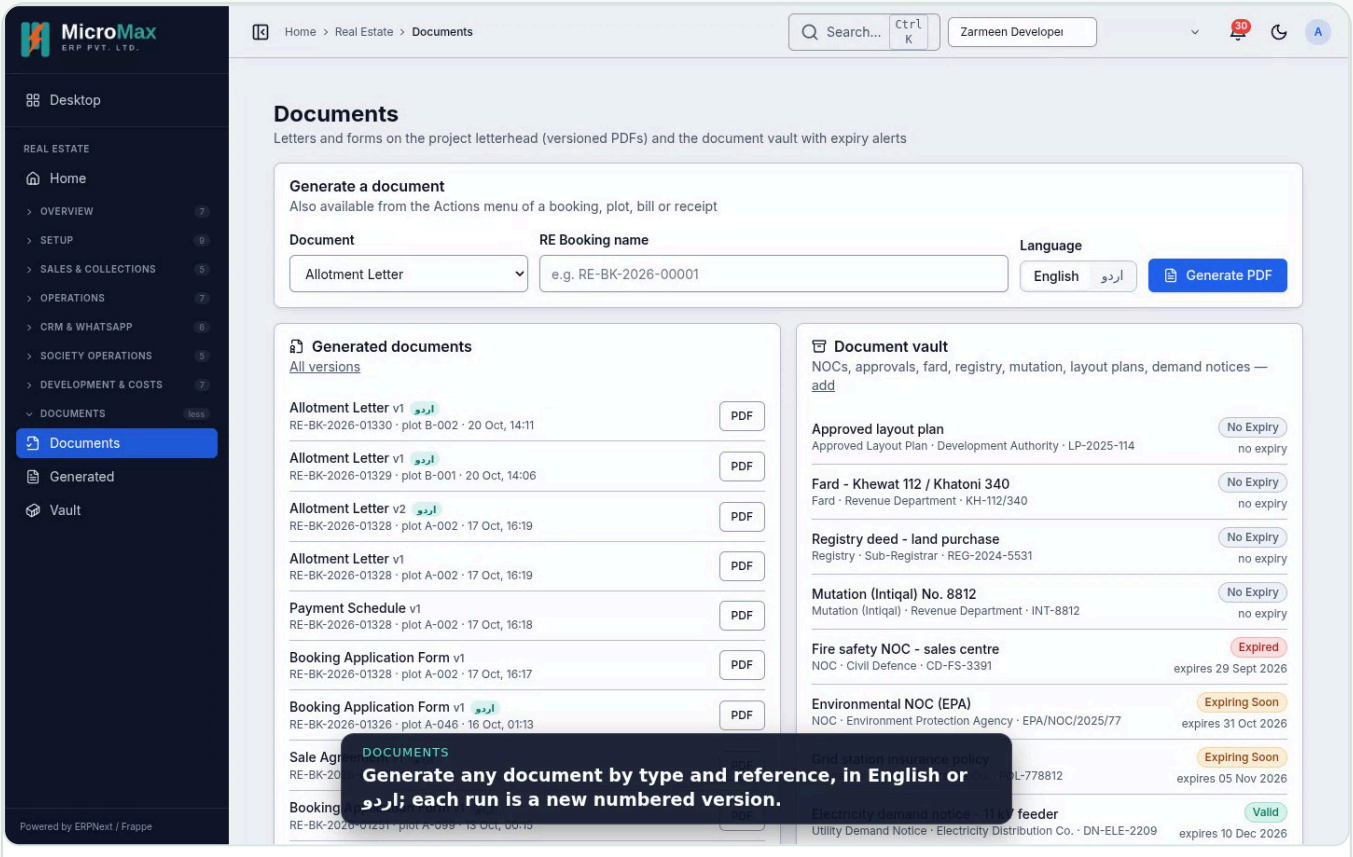
12 Maintenance & complaints

Plans, the urgent queue, complaint jobs and infrastructure.



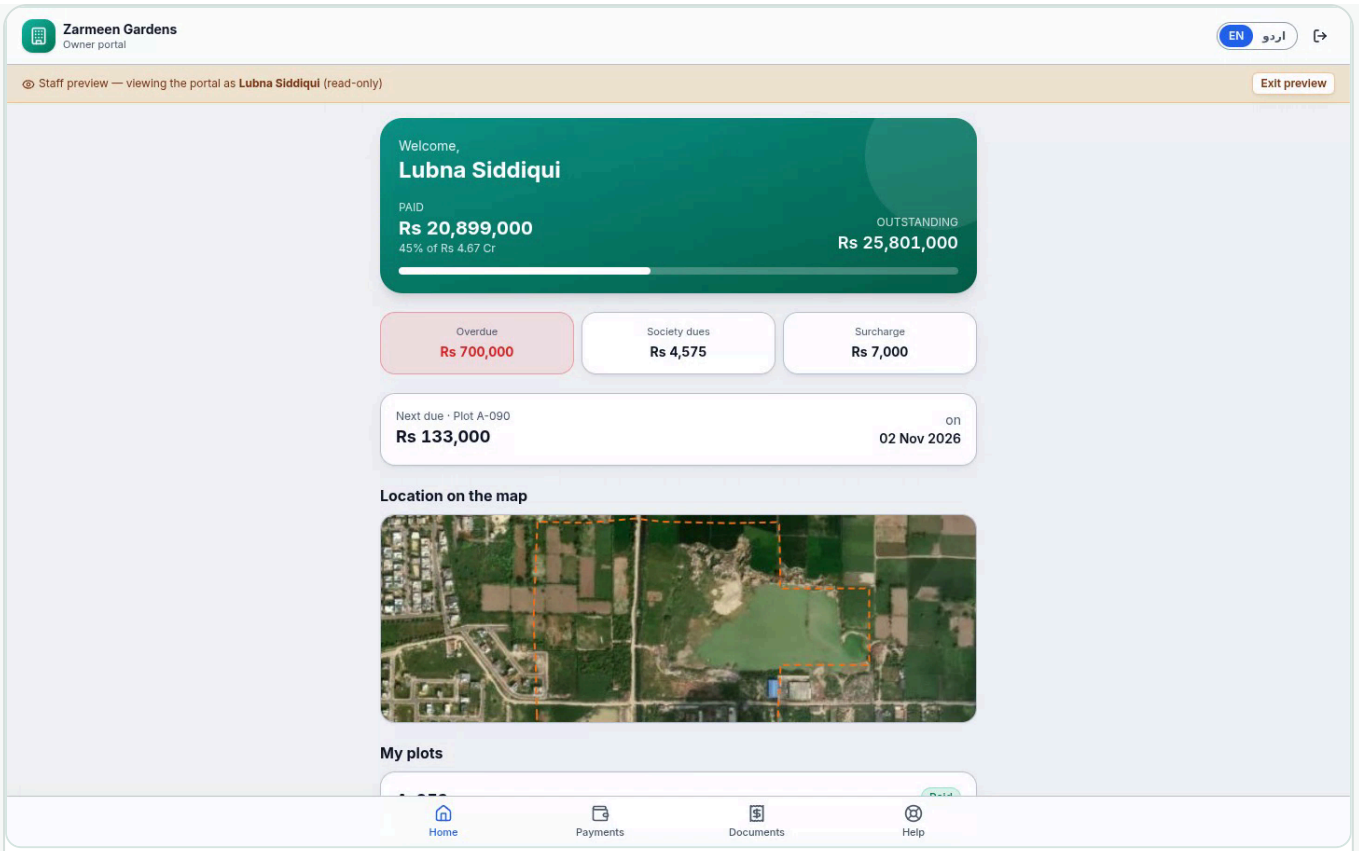
13 Development, stock & assets

Budget vs actual, procurement, material issues and assets with depreciation.



14 Documents & reports

Documents in English or Urdu and the report centre with charts.



15 Owner portal

What a plot owner sees, in English and Urdu.